



MOTOROLA

**MC10318L
MC10318L9**

Advance Specifications and Applications Information

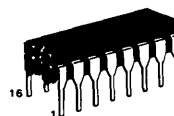
HIGH SPEED 8-BIT DIGITAL-TO-ANALOG CONVERTER

The MC10318 is a high speed 8-bit D/A converter capable of data conversion rates in excess of 25 MHz. It is intended for applications in high speed instrumentation and communication equipment, display processing, storage oscilloscopes, radar processing, and TV broadcast systems. The inputs are compatible with MECL 10,000 series logic, while the complementary current outputs have 51 mA full scale capability. 8-bit accurate ($\pm 1/2$ LSB) and monotonic over the full temperature range, the outputs typically settle in less than 10 ns.

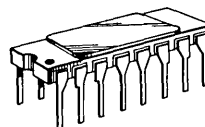
- FAST! Settling Time — 15 ns Typ
- 8-Bit Accuracy ($\pm 0.19\%$) — MC10318L
9-Bit Accuracy ($\pm 0.1\%$) — MC10318L9
- Inputs MECL 10,000 Compatible
- Complementary Current Outputs
- Output Compliance: -1.3 V to $+2.5$ V
- Standard: -5.2 V Supply
- Standard 16 Pin Ceramic Package
- Low Dissipation — Typically Less Than 500 mW
- Low Cost

HIGH SPEED 8-BIT DIGITAL-TO-ANALOG CONVERTER

**SILICON MONOLITHIC
INTEGRATED CIRCUIT**



**L SUFFIX
CERAMIC PACKAGE
CASE 620**

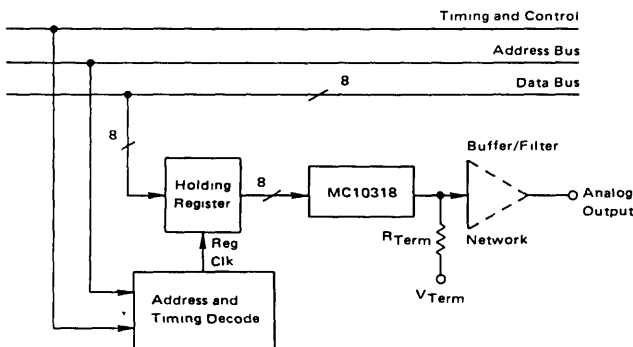


**L SUFFIX
CERAMIC PACKAGE
CASE 690**

Motorola reserves the right to supply this device in either of the above packages.

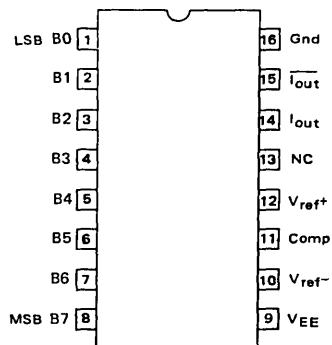
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TYPICAL MC10318 TO MC10800 PROCESSOR INTERFACE



This is advance information and specifications are subject to change without notice.

PIN CONNECTIONS



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MAXIMUM RATINGS (T_A = +25°C unless otherwise noted.)

Rating	Symbol	Value	Unit
Power Supply Voltage	V _{EE}	-6.0 to +0.5	Vdc
Digital Input Voltage	V _I	0 to V _{EE}	Vdc
Applied Output Voltage	V _O	+5.0	Vdc
Reference Current	I _{ref} (I ₂)	5.0	mA
Output Current	I _{FS}	75	mA
Reference Amplifier Input Range	V _{ref}	+0.5 to V _{EE}	Vdc
Reference Amplifier Differential Inputs	V _{ref} (D)	±5.0	Vdc
Operating Temperature Range	T _A	0 to +70	°C
Storage Temperature Range	T _{stg}	-65 to +150	°C
Junction Temperature Ceramic Package	T _J	+175	°C

CHARACTERISTICS

These specifications apply for V_{EE} = -5.2 V,
I_{FS} = 51 mA, T_A = 0°C to +70°C after thermal
equilibrium is reached.

@ Test
Temperature
0°C
25°C
70°C

TEST VOLTAGE VALUES (Note 1)				
Volts				
V _{IHmax}	V _{ILmin}	V _{IHAMin}	V _{ILAMax}	V _{EE}
-0.845	-1.868	-1.151	-1.516	-5.2
-0.810	-1.850	-1.105	-1.505	-5.2
-0.727	-1.830	-1.052	-1.480	-5.2

Characteristics	Symbol	Min	Typ	Max	Unit
Power Supply Voltage Range	V _{EE}	-5.46	-5.2	-4.94	V
Power Supply Current (Pins 1 thru 8 Open, I _{FS} = 51 mA)	I _{EE}	—	90	130	mA
Monotonicity	—	8.0	8.0	—	Bits
Nonlinearity MC10318L	—	—	—	±0.19	% FS
MC10318L9	—	—	—	±0.10	
Settling Time to 1/2 LSB (All Bits Switched On or Off, T _A = 25°C, Note 3)	t _s	—	15	—	ns
Full Scale Output Temperature Drift	TCI _{FS}	—	±50	±150	ppm/°C
Full Scale Current — Figure 1 (R3, R4 = 3.300 kΩ, V _{ref} = 10.560 V, Note 2)	I _{FS}	46.000	51	56.000	mA
Zero Scale Current (Note 2)	I _{ZS}	—	5.0	50	μA
Full Scale Symmetry (I _{FS15} - I _{FS14} , Note 2)	I _{FSS}	—	15	100	μA
Half Scale Accuracy MC10318L	HSA	—	—	±50	μA
(I _{HS} = 25.5 mA) MC10318L9	—	—	—	±25	
Output Voltage Compliance (Note 2) Full Scale Current Change ≤ 99 μA 50 μA	V _{OC}	-1.3	—	2.5	V
Power Supply Sensitivity (of Full Scale Current) (V _{EE} = -4.94 V to -5.46 V)	PSSI _{FS}	—	±0.002	±0.02	%/%
Reference Bias Current, Pin 10 (I _{ref} = 3.2 mA)	I ₁₀	—	6.0	15	μA
Propagation Delay 50% to 50% (All Bits Switched Low to High, High to Low)	t _p	—	3.0	—	ns

- NOTES. 1. Logic input levels are compatible with MECL 10,000 logic series.
2. Output characteristics apply to both pins 14 and 15, I_{out} and I_{out}.
3. See comments on construction and evaluation techniques in Figure 2 and text.

FIGURE 1 - FULL SCALE AND HALF SCALE CURRENT TEST CIRCUIT

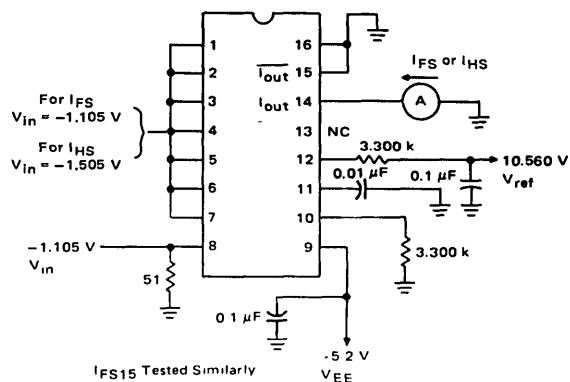
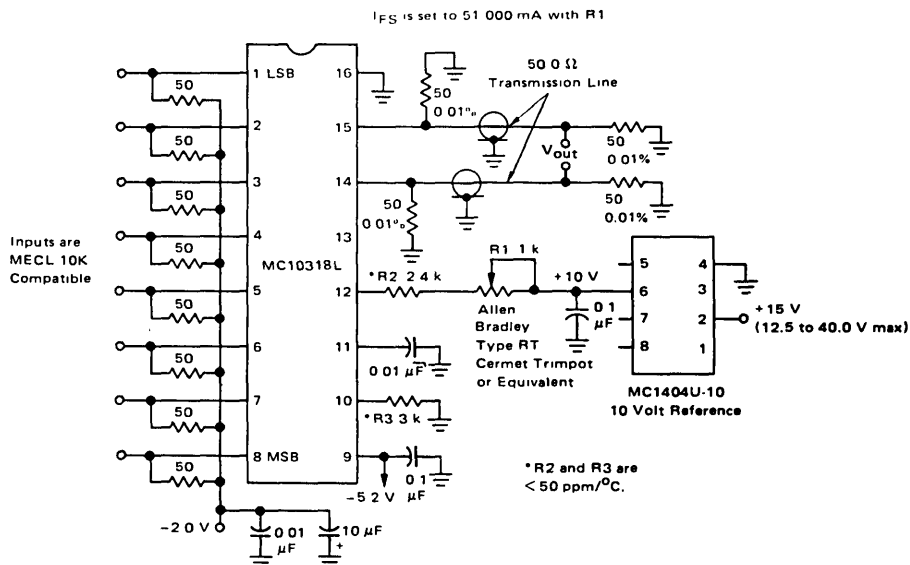


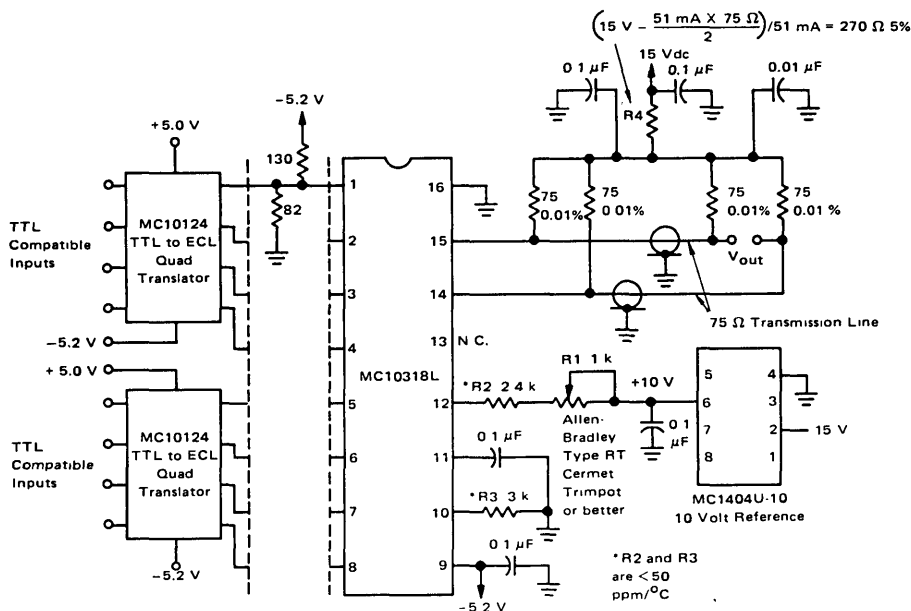
FIGURE 2 - TYPICAL CONNECTIONS FOR 50 Ω TRANSMISSION LINE



NOTE. Line impedances and termination impedance must be homogeneous 50.00 Ω. Any deviation will cause reflections which will seriously affect settling time. Optimum performance cannot be realized with sockets. Good 1.0 GHz microstripline techniques must be used.

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FIGURE 3 – TYPICAL CONNECTIONS FOR 75 Ω TRANSMISSION LINE AND TTL-COMPATIBLE INPUTS



NOTE: See caution on line and termination impedance in Figure 2 and text

APPLICATION INFORMATION

Functional Test Circuit Construction

Test circuits used to evaluate this device or circuit designs used in actual practical situations must employ good 1.0 GHz RF microstripline practices if optimum performance is to be achieved from this device. Both line and termination impedances must be matched to within $\pm 0.19\%$ to minimize reflections which will appear as increased settling time. The use of sockets for initial evaluation is not recommended if specified settling time is to be obtained.

Applications information can be obtained by contacting:

Application Engineering
(602) 244-3021

If desired, test circuit artwork and board specifications will be supplied by contacting:

Linear Interface Marketing
(602) 962-2294

Successive Approximation A/D Converter

The circuit shown in Figure 4 uses the MC10318 in a successive approximation analog-to-digital converter. The circuit as shown will operate at a clock frequency above 30 MHz if proper attention is given to layout.

The full-scale voltage (V_{FS}) for the circuit as shown is 10.20 V. This full-scale voltage may be changed by changing the 200 Ω resistor to a value given by:

$$R = \frac{V_{FS}}{I_{FS}} = \frac{V_{FS}}{51 \text{ mA}}$$

However, at low values of V_{FS} the resolution of the comparator must be considered to maintain a $\pm 1/2$ LSB accuracy.

FIGURE 4 – SUCCESSIVE APPROXIMATION A/D CONVERTER USING MC10318

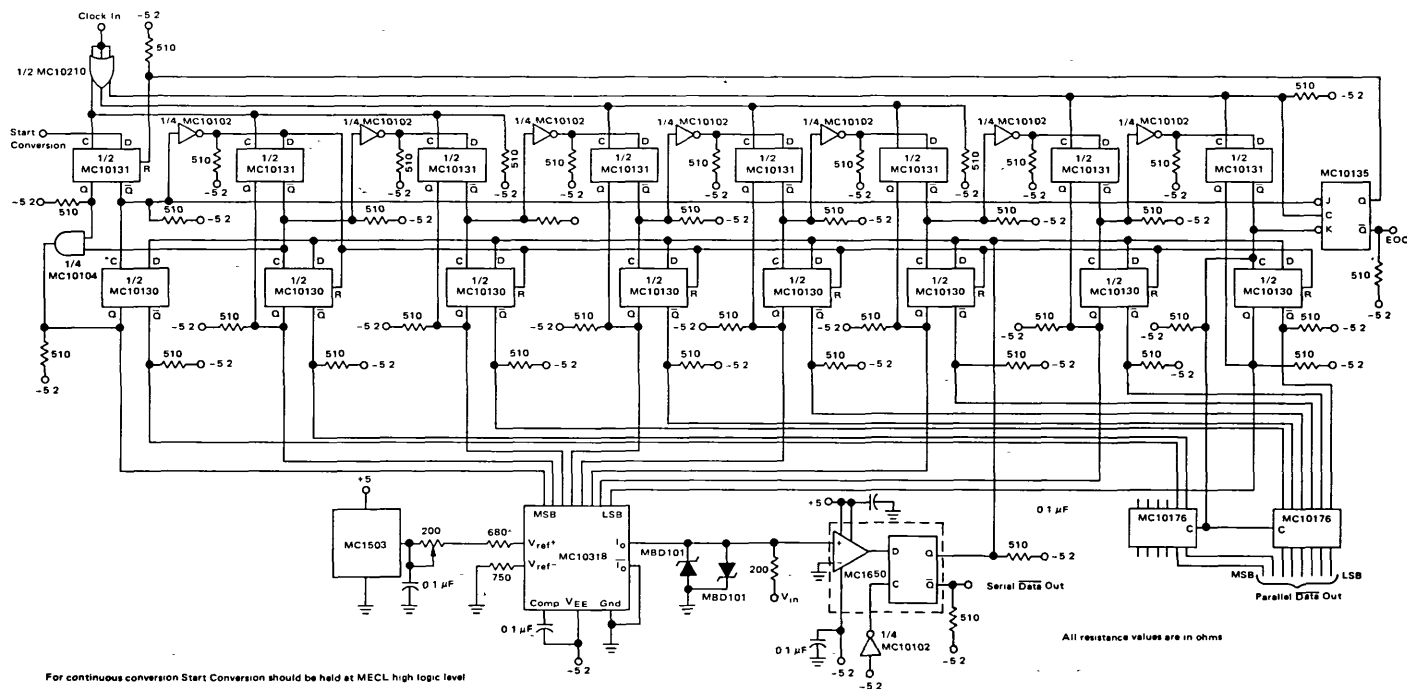


FIGURE 5 – MC10318 EQUIVALENT CIRCUIT

